

ADVANCED COMPUTER NETWORK PROTOCOLS

Comprehensive Study Notes

Targeted for: GATE | HPCL | ISRO | DRDO | Information Security Officer CBT

OSI Layer-wise Protocol Classification | Step-by-step Working | Flowcharts | Comparison Tables | Quick
Revision

Section 0 — OSI Model & Protocol Classification

The **OSI (Open Systems Interconnection)** model has 7 layers. Every network protocol operates at a specific layer. Understanding layers helps identify a protocol's role, scope, and interactions.

OSI Layer	Layer #	Key Function	Protocols
Physical	L1	Bit transmission over medium	Ethernet (PHY), DSL, Fiber
Data Link	L2	Framing, MAC, error detection	Ethernet, PPP, STP, VLAN, Token Ring
Network	L3	Logical addressing, routing	IP, ARP, RARP, ICMP, NAT, PAT
Transport	L4	End-to-end delivery, flow control	TCP, UDP, SCTP
Session	L5	Session management, sync	NetBIOS, PPTP, RPC, SDP, RTCP, PAP
Presentation	L6	Data format, encryption, compression	SSL, TLS, XDR, NDR, AFP, LPP, NCP
Application	L7	User-facing network services	HTTP, FTP, SMTP, DNS, DHCP, SSH, SNMP...

Section 1 — Data Link Layer (Layer 2) Protocols

1.1 Ethernet (IEEE 802.3)

Standard: The most widely used LAN technology. Defined by **IEEE 802.3**.

1.1.1 Ethernet Frame Structure

Ethernet Frame (IEEE 802.3)

Preamble	Dest MAC	Src MAC	Type	Payload	FCS
7 bytes	6 bytes	6 bytes	2 bytes	46-1500 bytes	4 bytes

Preamble: 7 bytes of 10101010 for synchronization

SFD (Start Frame Delimiter): 10101011 — signals start of frame

Dest/Src MAC: 48-bit hardware addresses

Type/Length: >1500 = EtherType (IPv4=0x0800, ARP=0x0806)

FCS: 4-byte CRC for error detection

1.1.2 CSMA/CD — How Ethernet Works

Ethernet uses **CSMA/CD** (Carrier Sense Multiple Access with Collision Detection) in half-duplex mode.

Ethernet CSMA/CD Flowchart

START: Host wants to send data

|

v

SENSE the medium — Is it idle?

|

YES

|

NO --> WAIT and retry

|

v

BEGIN transmission

|

v

MONITOR for collision

|

No Collision

|

COLLISION DETECTED

|

TRANSMISSION

|

Send JAM signal (32-bit)

COMPLETE

Stop transmission

|

Binary Exponential Backoff

Wait random time, retry

1.1.3 Key Facts for Exam

- Minimum frame size: **64 bytes** (to detect collisions)
- Maximum frame size (MTU): **1518 bytes** (1500 payload + 18 header)
- Duplex modes: **Half-duplex** (CSMA/CD applies), **Full-duplex** (no collision possible)
- Ethernet speeds: 10 Mbps → 100 Mbps (Fast) → 1 Gbps → 10 Gbps
- MAC address: 48 bits = 6 bytes; first 3 bytes = OUI (vendor), last 3 = device ID

1.2 Token Ring (IEEE 802.5) — High-Level

Attribute	Details
Standard	IEEE 802.5
Topology	Logical ring, physical star
Access Method	Token passing (no collisions possible)
Speed	4 Mbps or 16 Mbps
Key Feature	Deterministic — every device gets a turn (token)
Status	Obsolete — replaced by Ethernet

1.3 PPP — Point-to-Point Protocol

PPP is used for direct connections between two nodes (e.g., **dial-up**, **DSL**, **VPN**). It supports authentication, error detection, and multiple network protocols.

1.3.1 PPP Frame Structure

PPP Frame

Flag	Address	Control	Protocol	FCS
0x7E	0xFF	0x03	2 bytes	2-4B

Flag: 0x7E marks frame boundaries

Protocol: 0x0021 = IP, 0xC021 = LCP, 0xC023 = PAP, 0xC223 = CHAP

1.3.2 PPP Phases (Connection Establishment)

PPP Connection Phases

Phase 1: DEAD — No physical connection

|

Phase 2: ESTABLISH — LCP negotiation (Link Control Protocol)

| (MTU, compression, auth method agreed)

Phase 3: AUTHENTICATE — PAP or CHAP authentication

|

Phase 4: NETWORK — NCP negotiates IP (IPCP)

| (IP addresses assigned)

Phase 5: OPEN — Data transfer phase

|

Phase 6: TERMINATE — LCP terminates connection

1.3.3 PAP vs CHAP Authentication

Feature	PAP (Password Auth Protocol)	CHAP (Challenge Handshake Auth)
Security	Weak — plaintext password	Strong — uses MD5 hash
Exchanges	2-way (ID + Password)	3-way (Challenge, Response, Accept)
Replay Attack	Vulnerable	Protected (random challenge)
Re-auth	Only at connect time	Periodically re-authenticates
Exam Tip	Protocol 0xC023	Protocol 0xC223

1.4 STP — Spanning Tree Protocol (IEEE 802.1D)

STP prevents **Layer 2 loops** in redundant switched networks by blocking redundant paths and creating a loop-free tree topology.

1.4.1 STP Port States

State	Duration	Actions
Blocking	20 sec (Max Age)	Receives BPDUs only; no forwarding, no learning
Listening	15 sec (Forward Delay)	Participates in active topology; no data forwarding
Learning	15 sec (Forward Delay)	Learns MAC addresses; no data forwarding
Forwarding	Normal operation	Forwards data; learns MAC addresses
Disabled	Administratively down	No STP participation

1.4.2 STP Key Concepts

- **Root Bridge:** Bridge with lowest Bridge ID (Priority + MAC). All STP decisions stem from here.
- **Root Port:** Non-root bridge port closest to Root Bridge (lowest path cost).
- **Designated Port:** Best port on each segment to forward traffic toward root.
- **BPDUs** (Bridge Protocol Data Unit): Frames exchanged between switches to elect Root Bridge.
- **Convergence time:** ~30–50 seconds for STP. RSTP (802.1w) converges in < 1 second.

STP Election Process

- Step 1: All switches claim to be Root Bridge (broadcast BPDUs)
- Step 2: Switch with LOWEST Bridge ID wins (Priority 0-65535, default 32768)
- Step 3: All other switches find their Root Port (lowest cost path to Root)
- Step 4: On each segment, one Designated Port is elected
- Step 5: All other ports are BLOCKED (non-designated, non-root = blocking)

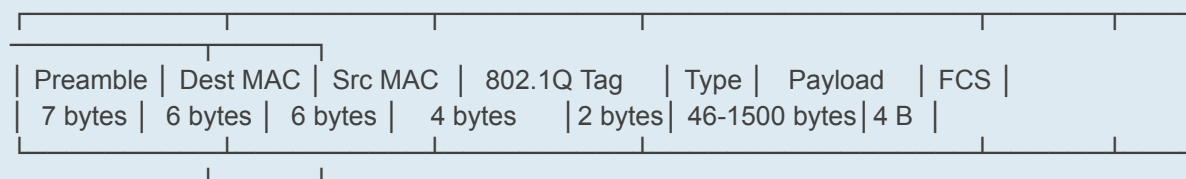
Path Cost: 100 Mbps = 19 | 1 Gbps = 4 | 10 Gbps = 2

1.5 VLAN — Virtual LAN (IEEE 802.1Q)

VLANs segment a physical network into multiple **logical networks** at Layer 2. Devices in different VLANs cannot communicate without a Layer 3 router.

1.5.1 802.1Q VLAN Tag

802.1Q Tagged Ethernet Frame



802.1Q Tag breakdown:

TPID (Tag Protocol ID): 0x8100

PCP (3 bits): Priority Code Point (0-7)

DEI (1 bit): Drop Eligible Indicator

VID (12 bits): VLAN ID → supports 4096 VLANs (0-4095, usable 1-4094)

1.5.2 VLAN Port Types

- **Access Port:** Connected to end devices. Carries traffic for ONE VLAN only. Untagged.
- **Trunk Port:** Connected between switches/routers. Carries tagged traffic for MULTIPLE VLANs.
- **Native VLAN:** Untagged traffic on a trunk port goes to Native VLAN (default: VLAN 1).
- **Inter-VLAN Routing:** Requires Layer 3 device. Done via Router-on-a-Stick or L3 switch.

Section 2 — Network Layer (Layer 3) Protocols

2.1 ARP — Address Resolution Protocol

ARP resolves a known **IP address** → **MAC address** for local network delivery. Without ARP, a host cannot construct an Ethernet frame even if it knows the destination IP.

2.1.1 ARP Packet Flow (Step-by-Step)

ARP Resolution Process

Scenario: Host A (192.168.1.10) wants to send data to Host B (192.168.1.20)

Step 1: Host A checks its ARP Cache

- If MAC for 192.168.1.20 found: use it directly
- If NOT found: send ARP Request

Step 2: ARP REQUEST (Broadcast)

Src IP: 192.168.1.10 | Src MAC: AA:BB:CC:11:22:33
Dst IP: 192.168.1.20 | Dst MAC: FF:FF:FF:FF:FF:FF (broadcast)
'Who has 192.168.1.20? Tell 192.168.1.10'

Step 3: All hosts receive broadcast; only 192.168.1.20 responds

Step 4: ARP REPLY (Unicast)

Src IP: 192.168.1.20 | Src MAC: DD:EE:FF:44:55:66
Dst IP: 192.168.1.10 | Dst MAC: AA:BB:CC:11:22:33
'I have 192.168.1.20! My MAC is DD:EE:FF:44:55:66'

Step 5: Host A stores mapping in ARP Cache (typically 20 min TTL)

Step 6: Host A sends the data frame to Host B's MAC

2.1.2 ARP Variants

Type	Purpose
ARP	IP → MAC (standard)
RARP (Reverse ARP)	MAC → IP (diskless workstations; obsolete, replaced by DHCP/BOOTP)
Proxy ARP	Router answers ARP on behalf of remote host (cross-network)
Gratuitous ARP	Host announces its own IP-MAC mapping (detect conflicts, update caches)
ARP Poisoning	Attack: send fake ARP replies to redirect traffic (MITM attack)

2.2 ICMP — Internet Control Message Protocol

ICMP (RFC 792) carries **error and diagnostic messages** for IP. It is NOT used for data transfer. Tools like **ping** and **traceroute** use ICMP.

2.2.1 ICMP Message Types (Critical for Exams)

Type	Code	Message	Use Case
0	0	Echo Reply	Ping response
3	0-15	Destination Unreachable	Host/port/net unreachable
5	0-3	Redirect	Router tells host of better route
8	0	Echo Request	Ping request
11	0	Time Exceeded (TTL=0)	Traceroute mechanism
12	0	Parameter Problem	Bad IP header

2.2.2 How Traceroute Uses ICMP

Traceroute Working

Step 1: Send UDP packet with TTL=1 → Router 1 drops it, sends ICMP 'Time Exceeded'
 Step 2: Send UDP packet with TTL=2 → Router 2 drops it, sends ICMP 'Time Exceeded'
 Step 3: ... (increment TTL each time, record responding router IP)
 Step N: Packet reaches destination → sends ICMP 'Port Unreachable' (Type 3, Code 3)

Result: Complete path from source to destination is mapped
 Windows tracert uses ICMP Echo; Linux traceroute uses UDP

2.3 NAT & PAT — Network/Port Address Translation

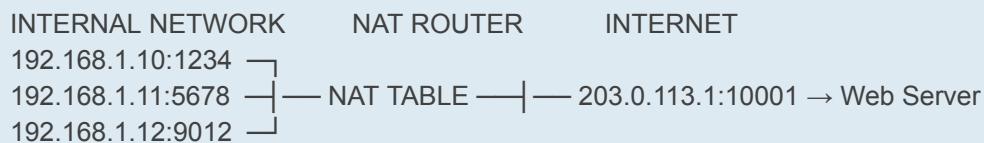
NAT allows **multiple private IP devices** to share one public IP. PAT (also called NAPT or NAT overload) extends this using **port numbers** to track individual connections.

2.3.1 Types of NAT

NAT Type	Mapping	Use Case
Static NAT	1 private IP ↔ 1 public IP (permanent)	Servers needing fixed public IP
Dynamic NAT	Private IP → pool of public IPs (temporary)	Multiple users, public IP pool
PAT / NAT Overload	Many private IPs → 1 public IP + port	Home/office routers (most common)

2.3.2 PAT — Step-by-Step Working

PAT (Port Address Translation) Example



NAT Translation Table:

Private IP:Port	Public IP:Port	
192.168.1.10:1234	203.0.113.1:10001	
192.168.1.11:5678	203.0.113.1:10002	
192.168.1.12:9012	203.0.113.1:10003	

Return traffic: Router matches port → forwards to correct private host

2.3.3 NAT vs PAT Comparison

Feature	NAT	PAT
Also Called	Basic NAT, Static/Dynamic NAT	NAPT, NAT Overload
Public IPs Required	Equal to or more than private	Only 1 public IP needed
Tracking Mechanism	IP address only	IP + Port number
Simultaneous Sessions	Limited (1:1 or pool)	Theoretically 65535 per IP
Most Common Use	Enterprise (server hosting)	Home routers, ISP networks

Section 3 — Session Layer (Layer 5) Protocols

Protocol	Full Name	Purpose	Key Feature
NetBIOS	Network Basic I/O System	Session mgmt for Windows LANs	Name resolution, session, datagram services over TCP/IP
ADSP	AppleTalk Data Stream Protocol	AppleTalk session protocol	Full-duplex byte stream; Apple legacy networks only
RTCP	Real-time Transport Control Protocol	Companion to RTP; QoS control	Sends SR/RR reports; monitors packet loss, jitter
PPTP	Point-to-Point Tunneling Protocol	VPN tunneling over IP networks	Encapsulates PPP frames; uses TCP port 1723; weak security
PAP	Password Authentication Protocol	Simple password authentication	Cleartext password; used in PPP; vulnerable to snooping
RPC/gRPC	Remote Procedure Call / Google RPC	Execute code on remote server	gRPC uses HTTP/2 + Protobuf; high performance microservices
SDP	Session Description Protocol	Describe multimedia sessions	Used with SIP/RTP; defines codec, IP, port for media
SCP	Session Control Protocol / Secure Copy	Protocol-level session control	SCP also = Secure Copy (SSH-based file transfer tool)

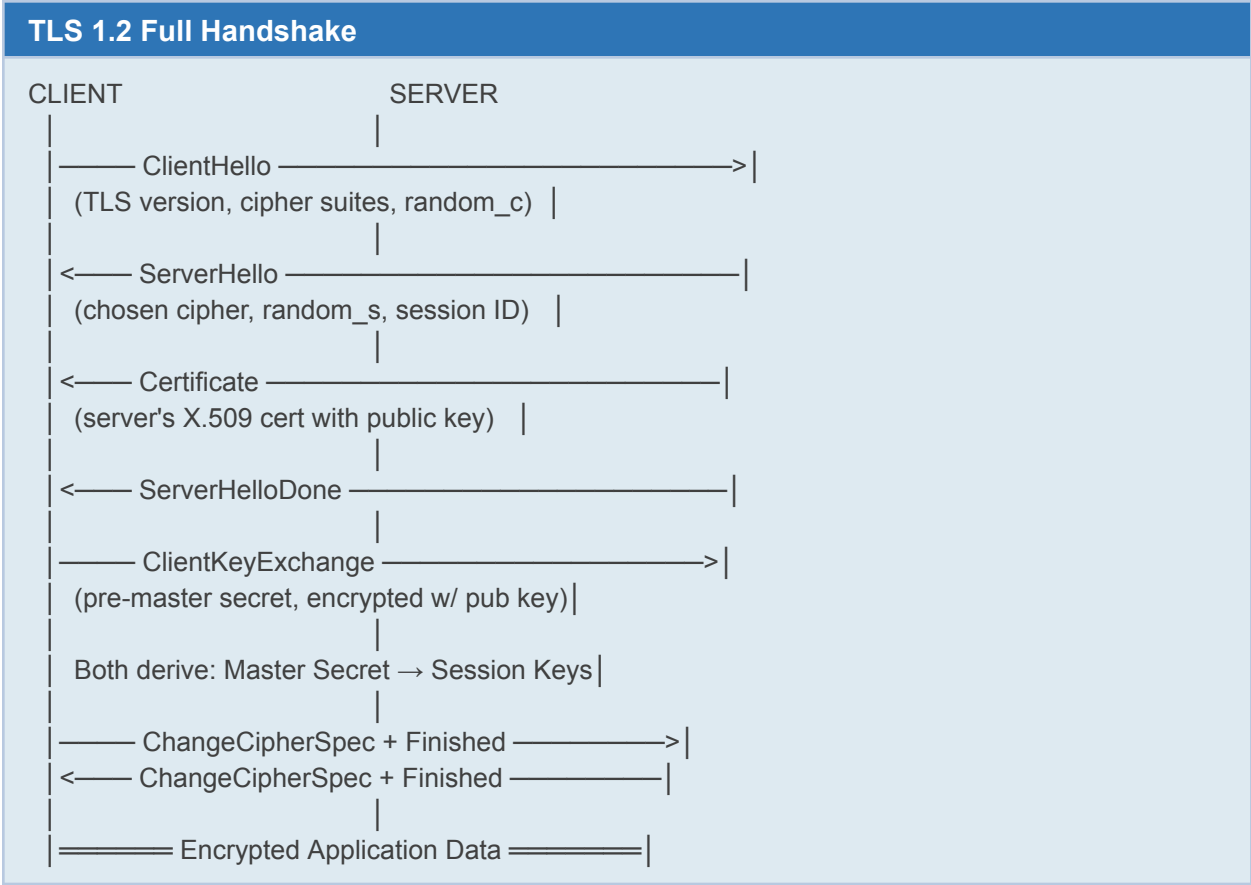
Section 4 — Presentation Layer (Layer 6) Protocols

Layer 6 handles **data formatting, encryption, decryption, and compression**. SSL and TLS are the most exam-important protocols at this layer.

4.1 SSL vs TLS — Handshake & Comparison

Feature	SSL (Secure Sockets Layer)	TLS (Transport Layer Security)
Developed by	Netscape (1994)	IETF (successor to SSL)
Versions	SSL 2.0, 3.0 (all deprecated)	TLS 1.0, 1.1 (deprecated), 1.2, 1.3
Current Status	Completely deprecated (insecure)	TLS 1.2 and 1.3 are current standard
Cipher Suites	Weak (RC4, MD5, DES)	Strong (AES-GCM, ChaCha20, SHA-256)
Handshake Rtt	2 RTT	TLS 1.3: 1 RTT (or 0-RTT for resumption)
Forward Secrecy	Not supported	Supported in TLS 1.3 (mandatory)
Port (HTTPS)	443 (historically)	443 (current standard)

4.1.1 TLS 1.2 Handshake (Step-by-Step)



4.1.2 TLS 1.3 Improvements

- **1-RTT handshake:** Reduced from 2-RTT (faster connections)
- **0-RTT resumption:** Session resumption without handshake (but replay attack risk)
- Removed weak ciphers: RC4, DES, 3DES, MD5, SHA-1 all removed
- **Forward Secrecy mandatory:** Uses ephemeral Diffie-Hellman (DHE/ECDHE)
- Encrypted by default: More of handshake is encrypted in 1.3

4.2 Presentation Layer — Other Protocols (High-Level)

Protocol	Full Name	Purpose / Key Feature
AFP	Apple Filing Protocol	Apple's file sharing protocol; uses port 548; replaced by SMB
LPP	Lightweight Presentation Protocol	Simplified ASN.1 encoding for OSI environments
NCP	NetWare Core Protocol	Novell NetWare file/print services; now largely obsolete
NDR	Network Data Representation	DCE/RPC data encoding; defines byte order, alignment
XDR	External Data Representation	Sun/NFS standard for platform-neutral data encoding; RFC 4506

Section 5 — Application Layer (Layer 7) Protocols

5.1 HTTP & HTTPS

HTTP (HyperText Transfer Protocol) is the foundation of web communication. **HTTPS** = HTTP over TLS — provides encryption, integrity, and authentication.

5.1.1 HTTP Methods

Method	Action	Idempotent?	Safe?
GET	Retrieve resource	Yes	Yes
POST	Submit data / create resource	No	No
PUT	Replace resource completely	Yes	No
PATCH	Partial update of resource	No	No
DELETE	Delete resource	Yes	No
HEAD	GET without body (headers only)	Yes	Yes
OPTIONS	Query supported methods	Yes	Yes

5.1.2 HTTP Status Codes

Range	Category	Examples
1xx	Informational	100 Continue, 101 Switching Protocols
2xx	Success	200 OK, 201 Created, 204 No Content
3xx	Redirection	301 Moved Permanently, 302 Found, 304 Not Modified
4xx	Client Error	400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found
5xx	Server Error	500 Internal Server Error, 502 Bad Gateway, 503 Service Unavailable

5.1.3 HTTP vs HTTPS Comparison

Feature	HTTP	HTTPS
Port	80	443
Security	No encryption (plaintext)	TLS encrypted
Certificate	Not required	SSL/TLS certificate required
Data Integrity	Not guaranteed	Guaranteed via MAC
Authentication	None	Server authenticated via cert; optionally client too
Performance	Slightly faster (no TLS overhead)	TLS 1.3 minimizes overhead
SEO	Penalized by Google	Preferred/required

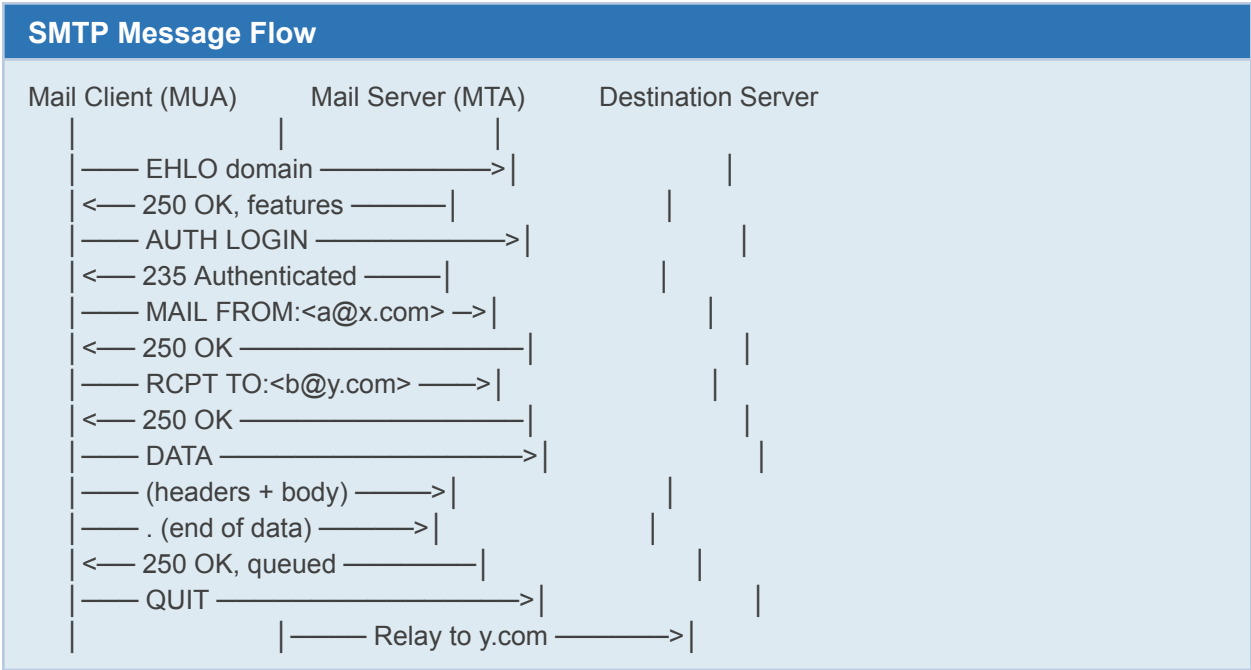
5.1.4 HTTP Versions

- **HTTP/1.0:** New TCP connection per request (slow)
- **HTTP/1.1:** Persistent connections, pipelining, Host header required
- **HTTP/2:** Multiplexing, header compression (HPACK), server push, binary framing
- **HTTP/3:** Uses QUIC (UDP-based), even lower latency, built-in TLS 1.3

5.2 Email Protocols — SMTP, POP3, IMAP

5.2.1 SMTP — Simple Mail Transfer Protocol

SMTP (RFC 5321) is used to **SEND** emails. It uses port **25** (server-to-server) or **587** (submission with auth). **SMTPS** uses port 465.



5.2.2 POP3 vs IMAP Comparison

Feature	POP3 (Post Office Protocol v3)	IMAP (Internet Message Access Protocol)
Port	110 (POP3), 995 (POP3S)	143 (IMAP), 993 (IMAPS)
Email Storage	Downloads & deletes from server	Emails stay on server
Multi-Device	Not suitable (emails on one device)	Ideal (synced across all devices)
Folder Support	No (flat inbox only)	Yes (server-side folders)
Offline Access	Yes (after download)	Limited (needs server connection)
Bandwidth	High (downloads all)	Low (downloads on demand)
State	Stateless after download	Stateful (tracks read/unread)

Feature	POP3 (Post Office Protocol v3)	IMAP (Internet Message Access Protocol)
Exam Tip	Download and delete model	Synchronization model

5.3 File Transfer Protocols — FTP, TFTP, SFTP

5.3.1 FTP — File Transfer Protocol

FTP (RFC 959) uses **two connections**: Control (port 21) and Data (port 20 in active mode). It is NOT encrypted — credentials are sent in plaintext.

5.3.2 FTP Active vs Passive Mode

Feature	Active Mode	Passive Mode
Data Connection Initiator	Server connects to client (port 20)	Client connects to server
Client Firewall	Often blocked (server initiates)	Usually works (client initiates all)
Control Port	21 (client → server)	21 (client → server)
Data Port	20 (server → client)	Server tells client which port to use
Use Case	When client has no firewall	Most common (NAT/firewall environments)

5.3.3 FTP vs SFTP vs TFTP Comparison

Feature	FTP	SFTP	TFTP
Port	21 (control), 20 (data)	22 (SSH port)	69 (UDP)
Transport	TCP	TCP (via SSH)	UDP
Encryption	None	Full (SSH)	None
Authentication	Username/password	SSH keys or password	None (anonymous)
Connections	2 (control + data)	1 (multiplexed in SSH)	1
Use Case	Legacy file transfer	Secure file transfer	Bootimg, firmware, TFTP server
Resume Transfer	Yes	Yes	No

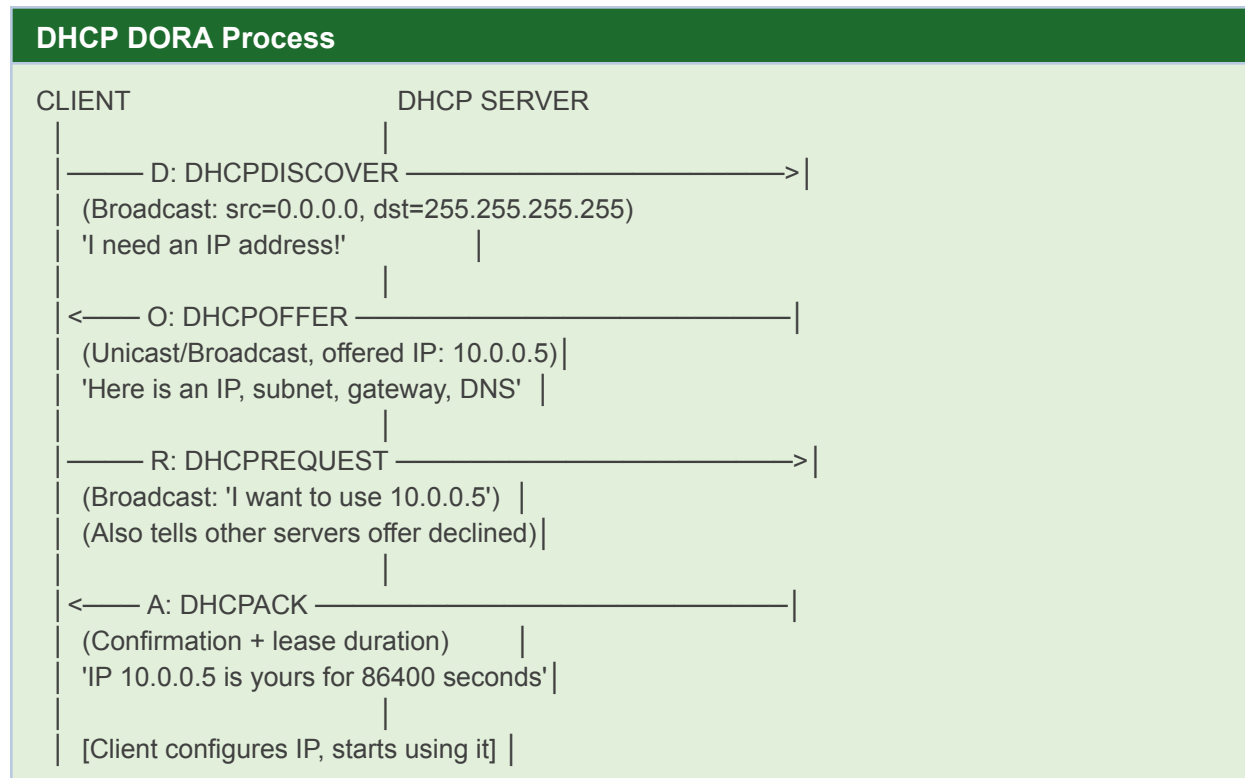
5.3.4 TFTP Specifics

- **Trivial FTP** — minimal protocol, no authentication, no directory listing
- Uses **UDP port 69** — connectionless, unreliable transport
- **Block size**: Default 512 bytes per block; uses stop-and-wait ARQ
- Primary uses: PXE boot (network booting), Cisco router IOS upload, TFTP servers in network labs

5.4 DHCP — Dynamic Host Configuration Protocol

DHCP (RFC 2131) automatically assigns **IP addresses and network configuration** to hosts. Port **67** (server) and **68** (client), UDP.

5.4.1 DORA Process — 4-Step IP Assignment



5.4.2 DHCP Lease Renewal

- At **T1 (50% of lease)**: Client sends DHCPREQUEST to renew (unicast to server)
- At **T2 (87.5% of lease)**: If no response, client broadcasts DHCPREQUEST to any server
- At **T3 (100% / expiry)**: Client must stop using IP and restart DORA
- **DHCPRELEASE**: Client voluntarily returns IP (e.g., on shutdown)
- **DHCPNACK**: Server rejects renewal (e.g., IP now belongs to different subnet)

5.4.3 DHCP Options (Key Fields in ACK)

Option	Code	Value Provided
Subnet Mask	1	e.g., 255.255.255.0
Router (Default Gateway)	3	e.g., 192.168.1.1
DNS Server	6	e.g., 8.8.8.8
Lease Time	51	e.g., 86400 (seconds)
DHCP Message Type	53	1=Discover, 2=Offer, 3=Request, 5=ACK, 6=NAK
DHCP Server ID	54	Server's IP address

5.5 DNS — Domain Name System

DNS (RFC 1035) translates **domain names** → **IP addresses**. It uses **UDP port 53** for queries (and TCP port 53 for zone transfers or large responses).

5.5.1 DNS Hierarchy

DNS Hierarchy (Top to Bottom)

ROOT DNS SERVERS (13 clusters worldwide, labeled a.root-servers.net to m.root-servers.net)

|

v

TLD (Top-Level Domain) Servers

.com | .org | .net | .in | .uk (managed by registries)

|

v

AUTHORITATIVE NAME SERVERS

ns1.example.com | ns2.example.com (managed by domain owner)

|

v

RESOURCE RECORDS

A: domain → IPv4 | AAAA: domain → IPv6

MX: Mail server | CNAME: Alias | NS: Name server | PTR: Reverse lookup | TXT: SPF/DKIM

5.5.2 DNS Resolution Process (Step-by-Step)

DNS Recursive Query Example: www.example.com → IP

Step 1: User types www.example.com in browser

Step 2: Browser checks its DNS cache → Not found

Step 3: OS checks /etc/hosts file → Not found

Step 4: Query sent to Recursive Resolver (usually your ISP or 8.8.8.8)

Step 5: Recursive Resolver checks its cache → Not found

Step 6: Resolver queries ROOT SERVER → 'Ask .com TLD server'

Step 7: Resolver queries .COM TLD SERVER → 'Ask ns1.example.com'

Step 8: Resolver queries AUTHORITATIVE SERVER (ns1.example.com)

→ Returns A record: www.example.com → 93.184.216.34

Step 9: Resolver caches result (TTL-based) and returns IP to client

Step 10: Browser connects to 93.184.216.34 via TCP/TLS

KEY: Recursive Resolver does the heavy lifting; client makes only ONE query

5.5.3 DNS Query Types

Query Type	Description	Who Uses It
Recursive	Resolver must return full answer or error	Client → Recursive Resolver
Iterative	Server returns best referral if can't answer	Resolver → Root/TLD/Auth servers
Inverse/Reverse	IP → Domain name (using PTR record in in-addr.arpa)	Email spam filtering, logging

5.6 Telnet vs SSH — Remote Access

Feature	Telnet	SSH (Secure Shell)
Port	23	22
Encryption	None — plaintext everything	Full encryption (AES, ChaCha20)
Authentication	Username/password (cleartext)	Password or public-key cryptography
Integrity	None	HMAC ensures data integrity
Man-in-Middle	Completely vulnerable	Protected via host key verification
Use Today	Legacy/lab equipment only	Standard for all remote admin
Data Tunneling	No	Yes — SSH tunneling/port forwarding
SFTP/SCP	No	Yes — file transfer built in

5.6.1 SSH Key-Based Authentication Flow

SSH Public-Key Authentication

SETUP (one time):

- Client generates key pair: id_rsa (private) + id_rsa.pub (public)
- Client's public key is added to server's ~/.ssh/authorized_keys

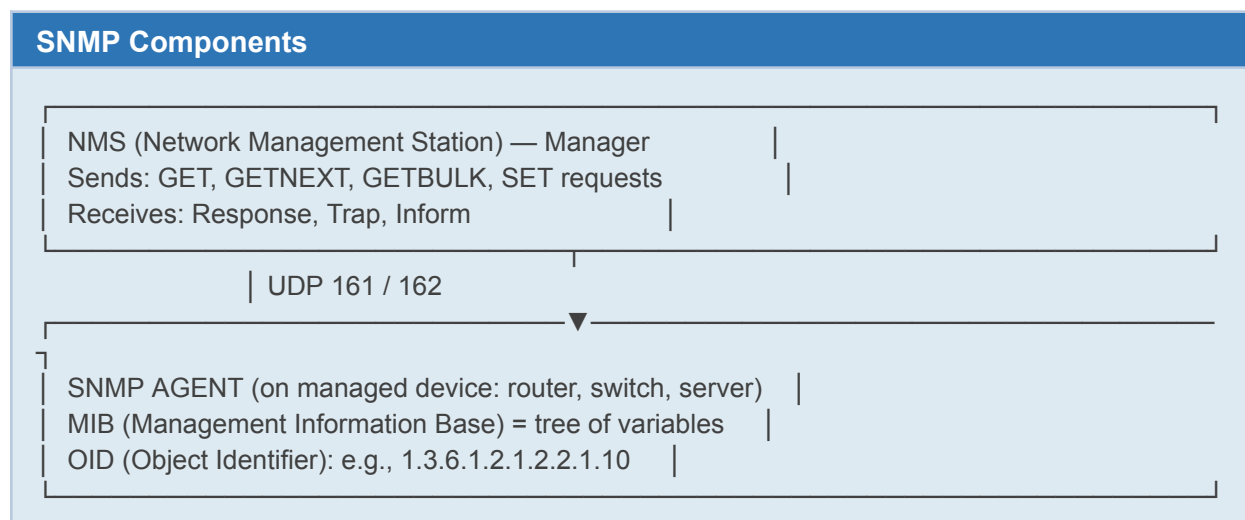
LOGIN PROCESS:

- Step 1: Client connects to SSH server (port 22), TCP handshake
- Step 2: Server sends its host key (client verifies against known_hosts)
- Step 3: Diffie-Hellman key exchange → shared session key
- Step 4: Server sends random challenge encrypted with client's public key
- Step 5: Client decrypts with private key → proves identity
- Step 6: Secure shell session established (all data encrypted)

5.7 SNMP — Simple Network Management Protocol

SNMP is used for **monitoring and managing network devices** (routers, switches, servers). Uses UDP port **161** (agent) and **162** (trap receiver).

5.7.1 SNMP Architecture

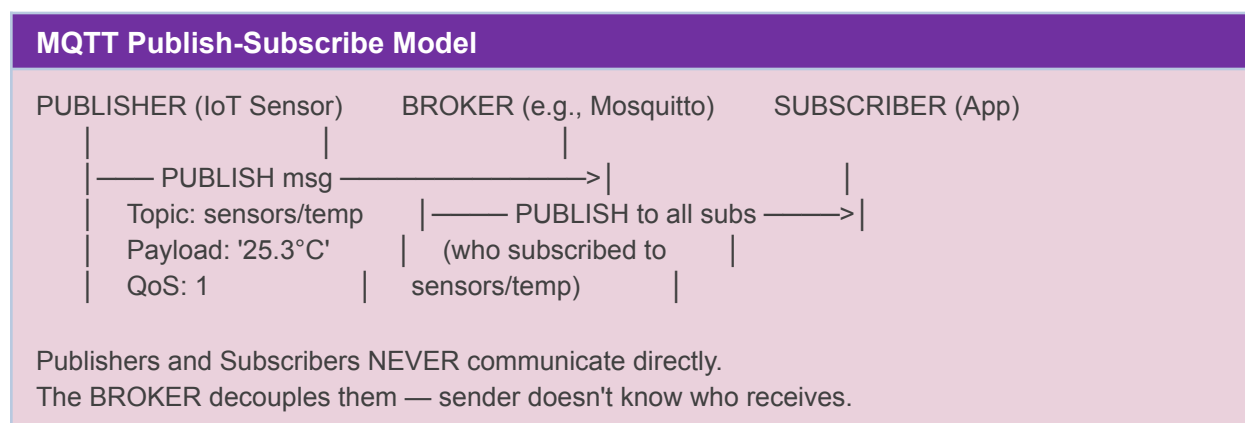


5.7.2 SNMP Versions

Version	Security	Key Feature
SNMPv1	Community string (cleartext)	Original; insecure; TRAP is unacknowledged
SNMPv2c	Community string (cleartext)	Adds GETBULK, INFORM (acknowledged trap)
SNMPv3	Authentication + Encryption	USM (auth with MD5/SHA) + PRIV (DES/AES); recommended

5.8 MQTT — Message Queuing Telemetry Transport

MQTT is a lightweight **publish-subscribe** messaging protocol designed for **IoT devices** with low bandwidth and high latency environments. Uses TCP port **1883** (or 8883 for TLS).

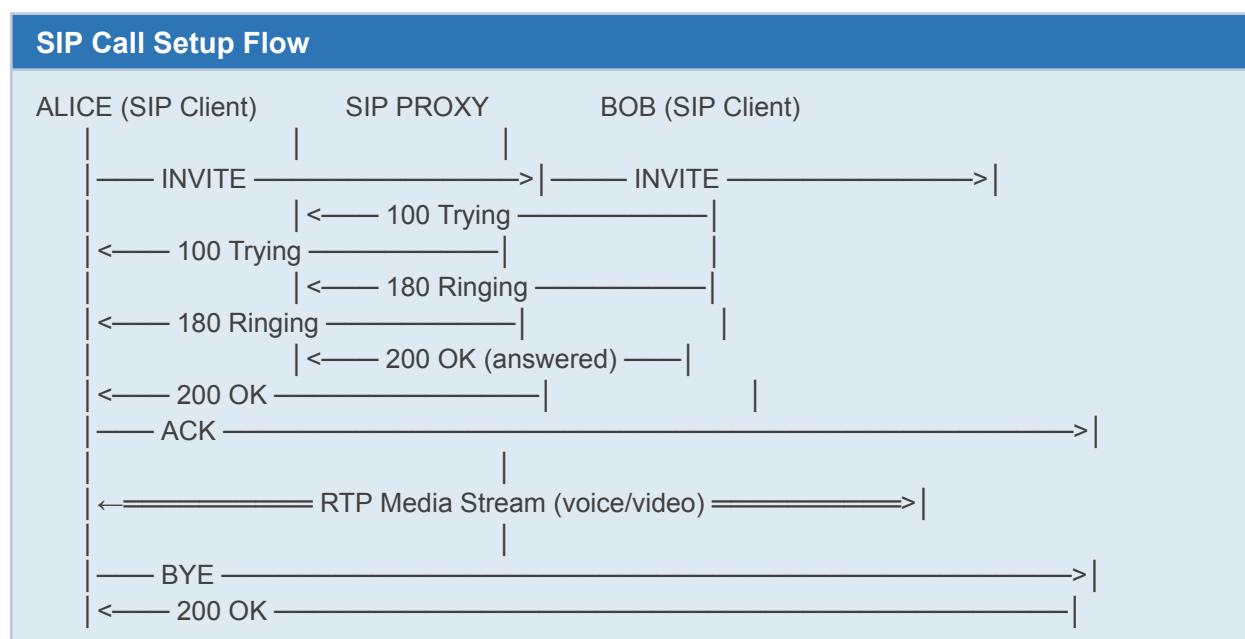


5.8.1 MQTT QoS Levels

QoS Level	Name	Delivery Guarantee	Use Case
QoS 0	At Most Once	Fire and forget — no ACK	Sensor data where loss is OK
QoS 1	At Least Once	Guaranteed delivery, possible duplicates	Most IoT messages
QoS 2	Exactly Once	4-way handshake, no duplicates	Critical commands

5.9 SIP — Session Initiation Protocol

SIP (RFC 3261) is used to **set up, modify, and terminate** multimedia sessions (VoIP calls, video conferencing). SIP is signaling-only; actual media uses **RTP**. Port **5060** (UDP/TCP) or **5061** (TLS).



Section 6 — Low/Medium Priority Protocols (High-Level Only)

These protocols require only basic awareness for competitive exams. Know: Layer, Purpose, and one distinguishing feature.

Protocol	Layer	Purpose	Key Feature / Exam Point
FTAM	L7	File Transfer & Management (OSI)	OSI-equivalent of FTP; rarely used; defined in ISO 8571
NFS	L7	Network File System (Unix/Linux)	Allows mounting remote directories locally; uses RPC; port 2049
LPD/LPR	L7	Line Printer Daemon/Remote	Print spooling over TCP port 515; predecessor to IPP
X Window	L7	GUI display across network	X11; client-server graphics; display server separate from apps
IRC	L7	Internet Relay Chat	Text-based group chat; port 6667; predecessor to Slack/Discord
MIME	L7	Multipurpose Internet Mail Extensions	Email attachments, HTML email; defines Content-Type headers
NNTP	L7	Network News Transfer Protocol	Usenet newsgroups; port 119; client-server discussion system
LDAP	L7	Lightweight Directory Access Protocol	Directory services (Active Directory); port 389 / 636 (LDAPS)
NTP	L7	Network Time Protocol	Clock synchronization; UDP port 123; stratum-based hierarchy
RARP	L3	Reverse ARP — MAC → IP	Obsolete; diskless workstations; replaced by DHCP/BOOTP
Token Ring	L2	IEEE 802.5 LAN	Token passing, deterministic, no collision; obsolete
ADSP	L5	AppleTalk Data Stream Protocol	Legacy Apple session-layer; full-duplex byte stream
NetBIOS	L5	Network Basic I/O System	Windows LAN services; name resolution, sessions; port 137-139
PPTP	L5	Point-to-Point Tunneling Protocol	Early VPN; TCP 1723; weak security; superseded by L2TP/IPsec
RPC/gRPC	L5	Remote Procedure Call	Execute remote functions; gRPC uses HTTP/2 + Protocol Buffers
SDP	L5	Session Description Protocol	Describes media sessions for SIP/RTP; codec, port, IP
RTCP	L5	Real-time Transport Control Protocol	Companion to RTP; QoS reports; SR/RR packets; port RTP+1
AFP	L6	Apple Filing Protocol	Apple file sharing; port 548; superseded by SMB on macOS

Protocol	Layer	Purpose	Key Feature / Exam Point
LPP	L6	Lightweight Presentation Protocol	Simplified OSI presentation encoding
NCP	L6	NetWare Core Protocol	Novell file/print services; largely obsolete
NDR	L6	Network Data Representation	DCE/RPC data encoding standard; defines endianness, alignment
XDR	L6	External Data Representation	Sun/NFS data encoding; RFC 4506; platform-neutral

Section 7 — Critical Port Numbers Reference

Port	Protocol	Transport	Use
20	FTP-Data	TCP	FTP active mode data transfer
21	FTP-Control	TCP	FTP command control channel
22	SSH / SFTP / SCP	TCP	Secure shell, secure file copy
23	Telnet	TCP	Remote terminal (unencrypted)
25	SMTP	TCP	Email sending (server-to-server)
53	DNS	UDP/TCP	Domain name resolution
67	DHCP Server	UDP	DHCP offers to clients
68	DHCP Client	UDP	DHCP requests from clients
69	TFTP	UDP	Trivial file transfer
80	HTTP	TCP	Web traffic (unencrypted)
110	POP3	TCP	Email retrieval (download model)
119	NNTP	TCP	Usenet news
123	NTP	UDP	Network time synchronization
143	IMAP	TCP	Email retrieval (sync model)
161	SNMP	UDP	Network management queries
162	SNMP Trap	UDP	Unsolicited SNMP notifications
389	LDAP	TCP/UDP	Directory services
443	HTTPS / TLS	TCP	Encrypted web traffic
465	SMTPS	TCP	SMTP over SSL (legacy)
514	Syslog	UDP	System log messages
515	LPD/LPR	TCP	Line printer daemon
548	AFP	TCP	Apple filing protocol
587	SMTP Submission	TCP	Email client to server (auth)
636	LDAPS	TCP	LDAP over TLS
993	IMAPS	TCP	IMAP over TLS
995	POP3S	TCP	POP3 over TLS
1723	PPTP	TCP	Point-to-point tunneling
1883	MQTT	TCP	IoT messaging (unencrypted)
2049	NFS	TCP/UDP	Network file system
5060	SIP	UDP/TCP	VoIP signaling
5061	SIPS	TCP	SIP over TLS
8883	MQTT/TLS	TCP	IoT messaging (encrypted)

Section 8 — QUICK REVISION SHEET

Use this section 24–48 hours before your exam for final revision.

8.1 Layer-by-Layer Protocol Memory Map

Layer	Remember These	Memory Hook
L2 Data Link	Ethernet, PPP, STP, VLAN, Token Ring	Every Protocol Selects Valid Tech
L3 Network	ARP, RARP, ICMP, NAT, PAT	ARP Resolves IPs Navigating Network
L5 Session	NetBIOS, PPTP, RPC, SDP, RTCP	Never Pass Rude Stuff Randomly
L6 Presentation	SSL, TLS, XDR, NDR, AFP	Secure Transport eXchanges Networks Awesomely
L7 Application	HTTP, FTP, SMTP, DNS, DHCP, SSH, SNMP, MQTT, SIP	Have Fast Service, DNS Delivers SSH, SNMP Monitors Smartly

8.2 Critical Comparisons — One-Line Facts

- **HTTP vs HTTPS:** HTTP = port 80, no encryption | HTTPS = port 443, TLS encrypted
- **FTP vs SFTP:** FTP = cleartext TCP 21/20 | SFTP = encrypted SSH port 22
- **TFTP:** UDP 69, no auth, no encryption — only for trusted networks (PXE boot, firmware)
- **POP3 vs IMAP:** POP3 = download+delete (port 110) | IMAP = sync on server (port 143)
- **Telnet vs SSH:** Telnet = cleartext port 23 (NEVER use) | SSH = encrypted port 22
- **SSL vs TLS:** SSL is DEAD (deprecated) | TLS 1.2/1.3 = current standard
- **NAT vs PAT:** NAT = IP mapping only | PAT = IP+Port mapping (one public IP for many)
- **STP vs RSTP:** STP (802.1D) = 30-50s convergence | RSTP (802.1w) = < 1 second
- **PAP vs CHAP:** PAP = 2-way, cleartext | CHAP = 3-way, MD5 hashed challenge
- **ARP vs RARP:** ARP: IP→MAC | RARP: MAC→IP (obsolete, replaced by DHCP)

8.3 Key Acronyms & Full Forms

Acronym	Full Form	Acronym	Full Form
CSMA/CD	Carrier Sense Multiple Access/Collision Detect	DORA	Discover, Offer, Request, Acknowledge
BPDU	Bridge Protocol Data Unit	OID	Object Identifier (SNMP)
MTU	Maximum Transmission Unit	MIB	Management Information Base
TLS	Transport Layer Security	FTAM	File Transfer, Access and Management
LDAP	Lightweight Directory Access Protocol	RTCP	Real-time Transport Control Protocol
SDP	Session Description Protocol	NCP	NetWare Core Protocol
XDR	External Data Representation	NDR	Network Data Representation

Acronym	Full Form	Acronym	Full Form
PPTP	Point-to-Point Tunneling Protocol	AFP	Apple Filing Protocol

8.4 Exam Traps — Common Mistakes

WATCH OUT FOR THESE!

1. RARP is NOT DHCP — RARP only gives IP, DHCP gives IP + gateway + DNS + more
2. ARP works at Layer 3 logically but uses Layer 2 (MAC) addressing
3. SSL is COMPLETELY deprecated — TLS is the correct term for modern use
4. FTP uses TWO connections (21 for control, 20 for data in active mode)
5. TFTP uses UDP (not TCP) — it is unreliable but simple
6. ICMP is Layer 3 (Network), NOT Transport layer
7. HTTPS uses port 443, not 80+443 — only 443
8. SMTP is for SENDING mail — POP3/IMAP are for RECEIVING mail
9. PAP sends password in PLAINTEXT — CHAP uses challenge-response with hash
10. DNS uses UDP for queries, TCP for zone transfers (and large responses > 512 bytes)
11. DHCP uses BROADCAST for Discover and Request (not unicast!)
12. STP Blocking state still RECEIVES BPDUs — it just doesn't forward data frames
13. SNMP v1 and v2c use community strings (cleartext) — only v3 is secure
14. MQTT is publish-subscribe via BROKER — publishers never contact subscribers directly
15. SIP is SIGNALING only — actual voice/video goes through RTP

8.5 Protocol Flashcard — Ports, Transport & Layer

Protocol	Port	Transport	Layer	Purpose (one line)
ARP	N/A	Ethernet broadcast	L3	IP → MAC resolution
ICMP	N/A	IP directly	L3	Error/diagnostic messages (ping, traceroute)
DHCP	67/68	UDP	L7	Auto IP assignment (DORA process)
DNS	53	UDP/TCP	L7	Domain name → IP address
HTTP	80	TCP	L7	Web page delivery (unencrypted)
HTTPS	443	TCP	L7	Encrypted web (HTTP over TLS)
FTP	21/20	TCP	L7	File transfer (unencrypted)
SFTP	22	TCP	L7	Encrypted file transfer via SSH
TFTP	69	UDP	L7	Simple file transfer, no auth
SMTP	25/587	TCP	L7	Send email
POP3	110/995	TCP	L7	Download email from server
IMAP	143/993	TCP	L7	Sync email on server
Telnet	23	TCP	L7	Remote terminal (cleartext)

Protocol	Port	Transport	Layer	Purpose (one line)
SSH	22	TCP	L7	Secure remote access
SNMP	161/162	UDP	L7	Network device monitoring
MQTT	1883/8883	TCP	L7	IoT publish-subscribe messaging
SIP	5060/5061	UDP/TCP	L7	VoIP call setup/teardown
NTP	123	UDP	L7	Time synchronization
LDAP	389/636	TCP	L7	Directory services
TLS/SSL	443	TCP	L6	Encryption/authentication layer